

EXAMPLE

DI In Code

FROM MANUAL WIRING TO A REAL FRAMEWORK

Manual DI — no framework

```
// composition root — one place that wires the whole app
PaymentGateway payment = new StripeGateway(apiKey);
Logger logger = new FileLogger("app.log");
OrderService orderService =
    new OrderService(payment, logger);
```

Nothing magic here — just one place ("the composition root") that constructs everything and passes it along. This is DI even without a framework.

Spring (Java) — annotation-driven

```
@Component
class StripeGateway implements PaymentGateway { /* ... */ }

@Component
class OrderService {
    private final PaymentGateway payment;

    @Autowired
    OrderService(PaymentGateway payment) { this.payment = payment; }
}
```

The framework scans for `@Component` classes, works out the dependency graph, and wires it for you. Same idea, automated.